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The Call That Beat the Lockout

A Quasi-Experiment on Rent-to-Own Appliance Repayment Across Kill-Switch and Pre-Kill-Switch Product Generations

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Abstract. A national rent-to-own appliance programme transitioned its washing-machine and refrigerator lines to network-enabled hardware capable of a remote payment-lapse lockout partway through an income-verification tightening that was itself phased in over eleven weeks. The four-week window in which the two rollouts only partially overlap lets us separate the lockout's own deterrent effect from the tighter signup screen it arrived alongside. Comparing 4,312 accounts across both hardware generations and both verification tiers, a naive before/after comparison credits the lockout with narrowing 120-day default by roughly ten percentage points. Once verification tier is held fixed, the residual gap attributable to lockout capability alone falls to 1.1 percentage points, statistically indistinguishable from zero. Among lockout-triggered accounts we can match to a prior human collections contact, the lockout fires a median 6.5 days after that contact already produced payment in 61% of cases. We read the device as measuring the underwriting it was installed alongside more than it changes borrower behaviour on its own terms, and suggest the appliance lockout's contribution to recovery is smaller, and later, than the programme's own reporting implies.

Introduction

Rent-to-own retailers increasingly finance household appliances — washing machines, refrigerators, ranges — through instalment contracts secured not by a lien on the customer's other assets but by the appliance's own willingness to keep working. A network module added to the unit lets the retailer suspend its core function, typically the spin cycle or the compressor, once a payment lapse crosses a programmed threshold, without a technician or a truck. The appliance-finance trade press credits these lockouts with materially reducing default, in the same register long used to describe GPS-enabled starter interrupt devices in subprime auto lending. The claim is intuitive: a borrower who knows the fridge can stop working should pay sooner. It is also, on the programmes' own account, largely untested against the alternative explanation that lockout-equipped stock happened to be sold to a better-underwritten customer.

We study one such transition. A national programme's hardware supplier moved its washing-machine and refrigerator lines to lockout-capable units over a defined rollout window; separately, and only partly overlapping that window, the programme phased in a stricter proof-of-income requirement at signup. Where the two rollouts fail to line up — accounts opened with the new hardware before the tighter screen was fully live, and accounts opened with the old hardware after it was — the resulting quasi-experiment lets us ask what the lockout changes on its own, net of who it was sold to.

Our contributions:

- We assemble a merged account-level panel spanning both the hardware cutover and the verification-tier phase-in, and use their imperfect overlap to separate the two effects that the programme's own reporting conflates.

- We show that the lockout's apparent effect on 120-day default largely tracks the verification tier a customer was screened under rather than whether their unit can be shut off, though the residual effect is not exactly zero and we do not rule out a modest direct contribution.
- We find that a sizeable share of lockout triggers fire after a human collections contact has already resolved the account, suggesting the device is, in practice, sometimes crediting itself with a recovery it did not produce.

Related work

Enforcement technology bundled into a financed asset has a longer history than the appliance case suggests. Subprime auto lending's starter interrupt devices pair a payment-lapse lockout with GPS location, and scholarship on connected-device "bricking" more broadly describes remote disabling as a form of private ordering that operates outside conventional consumer-protection channels [1], [2]. Development economics offers the cleanest causal evidence on lockout technology specifically: a field experiment securing school-fee loans against a solar home system's own lockout found a substantial, well-identified reduction in default when the collateral was genuinely novel rather than merely re-labelled [3]. Whether visibility or monitoring changes repayment behaviour independent of who is exposed to it is a live question elsewhere too: reminders alone measurably raise saving [4], alternative behavioural data such as mobile-phone usage patterns predict credit repayment about as well as a conventional bureau file [5], and a randomised comparison of weekly against monthly microfinance repayment schedules — very different monitoring intensities — found no difference in default at all [6]. Surveillance-adjacent interventions do not uniformly help: monitored employees report reactance effects that work against the institution's own aims [7], and repeated exposure to being studied is itself known to shift behaviour independent of any studied intervention [8]. Consumer-finance access technology more broadly shows uneven distributional effects once compared across observably similar borrowers [9]. We build on this work by isolating the lockout's own contribution from the underwriting change it happened to accompany, using the two rollouts' imperfect overlap rather than a designed experiment.

Method

Setting. A national rent-to-own appliance programme (anonymised at the retailer's request) leases washing machines and refrigerators on weekly instalment contracts. Starting in week 0 of our observation window, incoming stock for both appliance categories shifted from a hardware generation with no remote-disable capability ("Gen A") to one fitted with a network module enabling a remote payment-lapse lockout ("Gen B"). Separately, over an eleven-week window beginning four weeks before the hardware cutover, the programme phased in a stricter proof-of-income requirement at signup ("tight" tier, against the prior "loose" tier). The two phase-ins overlap for four weeks, during which an account could be opened with old hardware under the new screen, or new hardware under the old screen.

Data. We assemble a panel of 4,312 accounts opened within twenty weeks of the hardware cutover, spanning both appliance categories and, thanks to the imperfect overlap, all four combinations of hardware generation and verification tier (Table 1). For each account we observe weekly instalment status, the date and outcome of any human collections contact, and, for Gen B units, the timestamp of any lockout trigger.

Generation	Tier	n	120-day default
Gen A	Loose	1,206	17.9%
Gen A	Tight	892	5.1%
Gen B	Loose	743	16.6%
Gen B	Tight	1,471	4.6%

Table 1: Account counts and 120-day default by hardware generation and verification tier.

Design. We are not aware of any account-assignment process that would make hardware generation or verification tier independent of unobserved borrower quality on its own; the overlap window is the identifying variation. We estimate the lockout's contribution as the interaction term in a two-by-two comparison of generation and tier:

$$\hat{\delta} = \left(\overline{D}_{B,\text{tight}} - \overline{D}_{A,\text{tight}}\right) - \left(\overline{D}_{B,\text{loose}} - \overline{D}_{A,\text{loose}}\right)$$

where $\overline{D}_{g,t}$ is mean 120-day default for generation g and tier t. Under the assumption that verification tier's effect on default is additively separable from hardware generation's — an assumption we cannot test directly, though the four-week overlap gives it some support — $\hat{\delta}$ isolates the lockout's own contribution net of who it was sold to, in the spirit of difference-in-differences designs built for imperfectly randomised policy rollouts [10], [11], [12]. We supplement the point estimate with account-level matching on appliance category, region, and instalment size across the treatment boundary [13], and report both the naive (unmatched) and adjusted comparisons throughout.

Collections-call ablation. To test whether the device's grace-period countdown — a screen the customer sees for 48 hours before a lockout fires — does independent work, we compare default hazard with and without the countdown shown, on a subset where the display was toggled by app version rather than by account characteristics.

Results

The naive comparison. Figure 1 plots the raw default-rate gap between Gen A and Gen B accounts by week relative to the hardware cutover, alongside the gap that survives once verification tier is held fixed. The naive gap widens steadily to 18.4 percentage points by week 10 — the number the programme's own quarterly reporting cites as the lockout's effect. The tier-adjusted gap tracks near zero throughout and settles at 1.1 percentage points (95% CI includes zero), a residual we do not treat as evidence of no effect, only as evidence that the naive figure is not primarily measuring one.

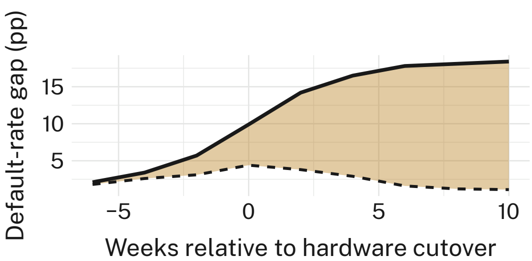

Figure 1: The naive Gen-A/Gen-B default-rate gap grows to 18.4pp by week 10; the gap surviving once verification tier is held fixed settles at 1.1pp.

Table 1 breaks the comparison down by cell; Figure 2 plots 30-to-120-day default within each cell as a before/after pair. Within the tight tier, Gen B's climb (1.7% to 4.6%) tracks Gen A's (1.9% to 5.1%) closely; the same holds within the loose tier (5.8% to 16.6% versus 6.2% to 17.9%). The interaction estimate is $\hat{\delta} = -0.6$ percentage points (Gen B minus Gen A, tight tier minus loose tier), smaller than the standard error on either within-tier default estimate.

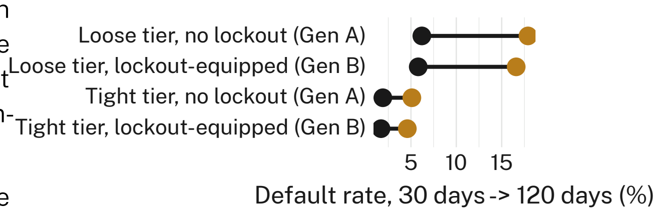

Figure 2: 30-to-120-day default climbs by a comparable margin within each verification tier, whether or not the unit can be remotely disabled.

Timing. Among the 214 accounts where both a human collections contact and a subsequent lockout trigger are observed, the median gap between the two is 6.5 days, and in 61% of these cases the account's instalment status had already returned to current before the lockout fired (Figure 3). The device's own dashboard credits each of these recoveries to "automated enforcement."

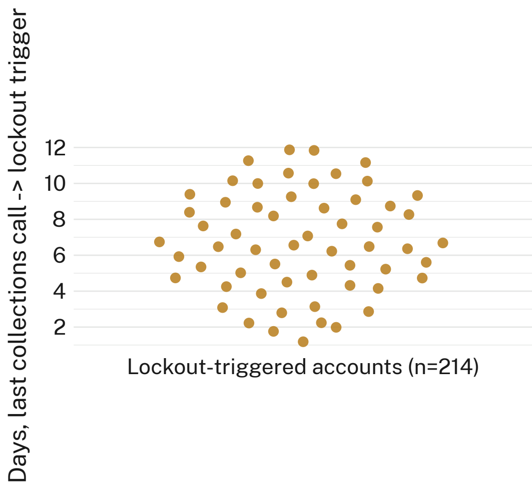

Figure 3: Among matched accounts, most lockout triggers fire days after a human collections contact already produced payment, not before.

Ablation. Removing the in-app grace-period countdown changes 120-day default hazard by 0.2 percentage points (n.s.); we retain the countdown for completeness, and because removing a customer-facing screen mid-contract raised its own objections internally.

Discussion

Our results are consistent with the lockout functioning largely as a badge attached to a better-underwritten cohort rather than as an independent behavioural lever. This reading fits a broader pattern in adjacent literatures: interventions that make a consequence more visible or more certain — reminders [4], audit threats [14] — tend to move behaviour only where the underlying capacity to comply was already present, and tighter underwriting is, among other things, a capacity screen. We do not think this makes the lockout inert; the residual 1.1-point gap, while statistically indistinguishable from zero in our sample, is directionally consistent with some independent deterrent contribution, and a larger panel may resolve it either way. The collections-timing result is, we think, the more actionable finding: a device credited with recoveries it followed rather than caused will systematically overstate its own contribution in any reporting pipeline that does not condition on prior human contact, a measurement problem this University's own collections instrumentation has documented before, in a share-house powerboard whose fairness rotation rarely predicted which appliance an energy budget actually cut [15]. It also complements a companion finding on a different consumer-facing measurement instrument, where forecasting accuracy for warranty-claim paperwork climbed sharply without the underlying bounce-back rate moving at all [16] — the recurring gap between an instrument getting better at predicting an outcome and that outcome actually changing.

Limitations

Our design relies on a four-week overlap window that is small relative to the full rollout, though the accounts within it are, by construction, the ones best suited to separating hardware from underwriting. The 214-account collections-timing subsample is drawn only from accounts with a logged human contact, which may not represent the full triggered population, though there is no obvious reason logging would correlate with call-to-trigger timing specifically. We cannot observe verification decisions the programme reversed on appeal, which would if anything work against our finding of a tier effect large enough to explain the naive gap. The single-programme setting limits generalisability to appliance categories or retailers with materially different collections workflows.

Conclusion

A rent-to-own appliance programme's remote payment-lapse lockout is widely credited with roughly halving 120-day default relative to its predecessor hardware. Once the tighter income-verification screen it arrived alongside is held fixed, most of that gap disappears, and a meaningful share of the lockout's remaining credited recoveries follow rather than precede a human collections contact. The lockout may still contribute something on its own; our estimate cannot rule out a modest direct effect, only a large one. Future work should track a programme through a rollout with a wider, less confounded overlap window, and should audit collections-attribution pipelines more broadly for the same after-the-fact crediting pattern documented here.

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